

# Connective Tissue, Vascular & Structural Findings

*A Summary of Objective Clinical Evidence Across 20+ Years of Records*

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## Purpose of This Document

This document organizes objective findings — imaging, laboratory results, and physician-documented observations — from over 20 years of medical records. Every item cited is sourced directly from clinical notes, imaging reports, or lab results. Nothing here represents patient self-report or self-diagnosis. The goal is to allow any provider to identify the documented structural and systemic pattern without reviewing the full record.

## Section 1: Documented Structural Pain Substrates

The following imaging findings establish an objective structural basis for musculoskeletal pain. These are confirmed radiologic findings, not clinical impressions.

### Tendon Pathology

| Location                | Finding                                                                                                                     | Source                                       |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Left Shoulder           | Partial bursal surface tearing of the supraspinatus tendon                                                                  | MRI                                          |
| Right Hip               | Partial gluteal tendon tearing                                                                                              | MRI (prior read)                             |
| Right Hip — Mayo Review | Greater trochanteric bursae trace edema/inflammation noted; gluteal tendons intact per 4/9/2026 Mayo re-read of outside MRI | MRI reinterpretation, Mayo Clinic 04/09/2026 |

**Note on MRI Discrepancy:** Prior outside reads documented partial gluteal tendon tearing. Mayo's 04/09/2026 reinterpretation of the same outside MRI (10/20/2023) describes tendons as intact with trace bursal edema. The discrepancy between reads of the same imaging warrants acknowledgment. Bursal inflammation was confirmed by both reads.

### Bone Lesions — Unexplained, Multi-site

| Location            | Finding                                                       | Source                    |
|---------------------|---------------------------------------------------------------|---------------------------|
| Left Iliac (Pelvis) | 7 mm hyperintense lesion                                      | MRI                       |
| T3 Vertebral Body   | 9 mm sclerotic focus, nonspecific                             | CT Chest, Mayo 04/15/2026 |
| Distal Left Femur   | 1.9 cm circumscribed sclerosis (nonossifying fibroma pattern) | MRI                       |

*Multi-site unexplained bone lesions in a patient with chronic eosinophilia and suspected systemic infiltrative disease warrant consideration of eosinophilic bone marrow involvement, which is a recognized manifestation of hypereosinophilic syndrome.*

## Hypermobility — Formally Documented

|                              |                                                                                                                                                     |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Diagnosis</b>             | Hypermobility Joint — Active Problem, Mayo Clinic (noted 04/01/2026)                                                                                |
| <b>Beighton Markers</b>      | Knee hyperextension, ankle hyperextension, thumb to wrist apposition — confirmed on physical exam                                                   |
| <b>Examining Physician</b>   | Dr. Maria Poirier, M.D., Mayo Clinic                                                                                                                |
| <b>Clinical Significance</b> | Hypermobility-associated connective tissue laxity is a recognized driver of chronic musculoskeletal pain, tendon instability, and joint dysfunction |

## Section 2: Vascular Anomalies & Connective Tissue Pattern

The following structural findings form a pattern consistent with a heritable connective tissue disorder. Each finding is individually documented in imaging reports; their co-occurrence is clinically significant.

### Vascular Anomalies

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- Aberrant right subclavian artery (retroesophageal origin from thoracic aorta) — confirmed CT Chest, Mayo 04/15/2026
- Few right apical blebs — CT Chest, Mayo 04/15/2026
  - Apical blebs in patients under 50 are strongly associated with heritable connective tissue disorders including Marfan syndrome, vEDS, and hEDS
- Mild diffuse bronchial wall thickening with patchy bilateral air trapping — CT Chest, Mayo 04/15/2026
- Right paratracheal stripe prominence — Chest X-ray 04/01/2026; CT recommended and performed

### Cardiac Structural Findings

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- Redundant interatrial septum — Echocardiogram 07/2024
  - Classic finding in hypermobility disorders; reflects tissue laxity
- Mild mitral valve regurgitation — Echocardiogram
- Mild tricuspid valve regurgitation — Echocardiogram
- Mild aortic valve regurgitation — Echocardiogram
  - Multivalvular regurgitation in a young patient without rheumatic history is a recognized connective tissue disorder finding
- Nonischemic cardiomyopathy (TTN pathogenic variant) — confirmed, active diagnosis
  - EF trend: 56% → ~50% over monitoring period
  - NT-proBNP 263 pg/mL (H, ref ≤162) — Mayo 04/01/2026

### GI & Pelvic Connective Tissue Manifestations

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Connective tissue disorders characteristically affect smooth muscle and pelvic floor integrity. The following are documented diagnoses:

- Irritable Bowel Syndrome — longstanding, documented
- Spastic pelvic floor syndrome (high-tone pelvic floor dysfunction)
- Internal and external hemorrhoids — documented, surgically treated 2026
- Trigonitis Bladder — Active Problem, Mayo Clinic (noted 05/01/2026)
- 3 mm bladder filling defect — CT Urogram, Mayo 04/01/2026; unchanged since 2016
- Prior hysterectomy and bilateral salpingo-oophorectomy — documented

## Section 3: TTN Variant — Musculoskeletal Implications

The confirmed pathogenic TTN variant (c.73318\_73324delGAATGCT, p.E24440MfsX2, heterozygous) has been evaluated not only for its cardiac effects but for potential musculoskeletal implications. Titin is expressed in both cardiac and skeletal muscle.

### Neurologist-Documented Assessment

*We discussed that TTN mutations can be associated with different presentations of muscular dystrophy, including distal myopathy, limb girdle muscular dystrophy.*

— Dr. Fernandes, UNMC Neurology, 2020 — documented in clinical note

- Chronic myalgia dating to high school — documented in neurological consultation
- Diffuse musculoskeletal pain across hips, lower back, ribs, sternum, and shoulders — documented as Chronic Pain Syndrome, active diagnosis
- Formal genetic consultation ordered by treating physician due to musculoskeletal concerns related to the TTN variant

*I have ordered genetic consultation as I am concerned that in addition to the cardiomyopathy, there are musculoskeletal issues related to this genetic pathogenic variant.*

— Dr. Maria Poirier, M.D., Mayo Clinic — documented in clinical note

### Family History Context

|                |                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------|
| <b>Father</b>  | Carried TTN pathogenic variant; developed progressive systolic heart failure; died from disease |
| <b>Brother</b> | Carries TTN pathogenic variant; further along in disease progression than patient               |
| <b>Pattern</b> | High penetrance; genetic anticipation pattern suspected across generations                      |

## Section 4: Systemic Inflammatory Context

The structural findings above occur in the context of an active, incompletely characterized systemic inflammatory process. The following is a summary of the inflammatory workup completed to date.

### Eosinophilia — Persistent, Documented

| Date            | Finding                                    | Source        |
|-----------------|--------------------------------------------|---------------|
| 04/01/2026      | Eosinophils 0.79 (H); Manual smear 14% (H) | Mayo Clinic   |
| 04/09/2026      | Eosinophils 0.58 (H); Manual smear 12% (H) | Mayo Clinic   |
| 12/23/2025      | Eosinophils 15% (H)                        | Broadlawns    |
| Historical peak | 15.2%                                      | Prior records |

### Exclusion Workup — Completed at Mayo Clinic April 2026

**The following conditions have been formally excluded:**

- **ANCA-positive vasculitis:** MPO <0.2, PR3 <0.2 — both negative
- **Systemic mastocytosis:** Tryptase 2.3 ng/mL (ref <11.5) — normal
- **B-cell lymphoproliferative disorder:** Flow cytometry: no monocytic B-cell population
- **T-cell lymphoproliferative disorder:** TCR PCR: no clonal T-cell rearrangement
- **Autoimmune CTD:** ANA <1:80 — negative
- **FIP1L1-PDGFRA positive CEL:** Chronic Eosinophilia FISH panel: all loci normal
- **Parasitic infection:** GI pathogen panel and O&P: all negative

Current Mayo working conclusion: Steroid-resistant eosinophilic asthma. Active treatment with mepolizumab (Nucala, initiated 04/16/2026). Differential not yet formally closed between ANCA-negative EGPA and idiopathic hypereosinophilic syndrome.

### Pulmonary Findings — April 2026

| Test             | Finding                                                                              | Date/Source         |
|------------------|--------------------------------------------------------------------------------------|---------------------|
| FeNO             | 82 ppb (ULN 39) — more than double upper limit                                       | 04/15/2026 Mayo     |
| Pre-BD FEV1/FVC  | 57.39% — obstructive pattern                                                         | PFT 04/15/2026 Mayo |
| Post-BD FEV1/FVC | 73.70% — significant bronchodilator response (+16 points)                            | PFT 04/15/2026 Mayo |
| CT Chest         | Bronchial wall thickening, bilateral air trapping, apical blebs, aberrant subclavian | 04/15/2026 Mayo     |



## Section 5: Pain Management Considerations

This section is intended to provide clinical context relevant to pain management planning given the patient's multi-system diagnosis profile.

### Current Pain Status

- Active diagnosis: Chronic Pain Syndrome — documented Mayo Clinic
- Pain distribution: hips, lower back, ribs, sternum, shoulders — documented across multiple provider notes
- Structural contributors confirmed on imaging: supraspinatus tearing, gluteal bursal inflammation, multi-site bone lesions, hypermobility with Beighton markers
- Myalgia history dating to high school — neurologist-documented

### Medication Constraints — Cardiac & Renal

#### NSAID Contraindication Context

NSAIDs (including ibuprofen and meloxicam) carry meaningful risk in this patient given:

- Nonischemic cardiomyopathy with declining EF — NSAIDs increase cardiac preload/afterload
- Baseline eGFR variability (documented 73 in 2025) — NSAIDs cause renal vasoconstriction
- Diastolic hypertension despite maximum beta blockade — NSAIDs blunt antihypertensive effect
- Elevated NT-proBNP (263, flagged abnormal) — fluid retention risk is additive
- Recent episode: meloxicam discontinuation without bridge → daily OTC ibuprofen use → acute bilateral ankle edema and BP elevation (June 2026)

### Steroid Dependency Pattern

- Multiple courses of oral prednisone for asthma and inflammatory flares — documented
- Steroid frequency has accelerated; disease is breaking through between courses
- December 2025 steroid course followed days later by worst recorded physiological crisis across HRV, eosinophil count, and inflammatory markers
- Post-steroid rebound pattern suggests inadequate underlying disease control rather than resolved condition

### Current Bridge Gap

#### As of June 2026:

Meloxicam was discontinued without notification or bridge plan. OTC ibuprofen was used in the interim and has now been stopped due to acute fluid retention. Acetaminophen alone is not providing adequate pain control. Next pain management appointment: June 18, 2026.

## Active Problem List — Mayo Clinic (as of May 29, 2026)

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| Diagnosis                                       | Noted      | System          |
|-------------------------------------------------|------------|-----------------|
| Cardiomyopathy in Diseases Classified Elsewhere | 04/01/2026 | Cardiac         |
| Cardiomyopathy Family History                   | 04/01/2026 | Cardiac/Genetic |
| Hypermobility Joint                             | 04/01/2026 | Musculoskeletal |
| Eosinophilia Unspecified                        | 04/01/2026 | Hematologic     |
| Asthma Severe Persistent                        | 04/01/2026 | Pulmonary       |
| Chronic Pain Syndrome                           | 04/01/2026 | Pain            |
| Tachycardia                                     | 04/01/2026 | Cardiac         |
| Posttraumatic Stress Disorder Prolonged         | 04/01/2026 | Psychiatric     |
| Allergy Pollen                                  | 04/01/2026 | Allergic        |
| Genetic Carrier of Other Disease                | 04/01/2026 | Genetic         |
| Mass Bladder                                    | 04/01/2026 | Urologic        |
| Trigonitis Bladder                              | 05/01/2026 | Urologic        |

*All findings sourced from clinical records | Prepared June 2026 | Not a self-diagnosis*